PH-214 Modern Physics Notes

Useful Math

Taylor series expansions:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots$$

$$(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n = 1 + mx + \frac{m(m-1)}{2}x^2 + \frac{m(m-1)(m-2)}{6}x^3 + \cdots$$

Note: For small x, higher order terms reduce to zero

Euler's Identity:
Use complex exponentials to manipulate complicated trig functions and complex amplitudes.

$$e^{ix} = \cos x + i \sin x$$

Visualizing propagating plane waves:
A 1D plane wave propagating in the $+z$ direction can be mathematically represented as:

$$\psi = f(kz - \omega t) \iff f(z - vt),\ v = \frac{\omega}{k}$$

Where:

- $(kz - \omega t)$ [rad]:
  The argument represents the **phase** of the wave. A change in phase simply moves the wave forward or backward in space and time, without changing its shape, there are a few ways to see this which we will look at later.

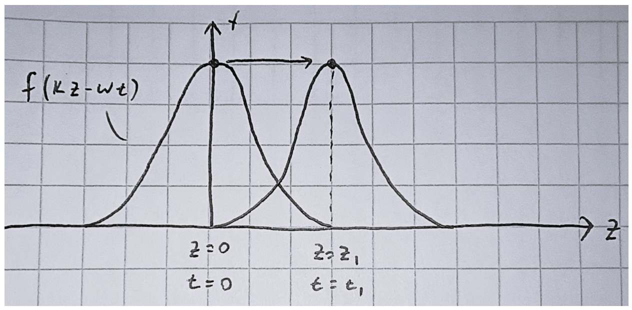
- k [rad/m]:
  is the **wave number**, representing the spatial frequency of the wave, or how *spread out* the wave is in space. Since k scales the spatial variable z, a larger k means the wave advances through the same phase over a smaller change in z.

- ω [rad/s]:
  is the **angular frequency**, representing the temporal frequency of the wave, or how *spread out* the wave is in time. As in rotational kinematics, ω scales time t, and a larger ω means the wave advances through the same phase over a smaller change in t.

A plane wave is special in that it represents a wave whose shape is fixed as it propagates through space and time, with it's shape represented by the function f.

Global View

As mentioned before, there are two ways to visualize how the phase $(kz - \omega t)$ represents a propagating wave. The first is a *global* view, looking at how z and t change relative to each other for a fixed phase, tracking a specific point on the wave.



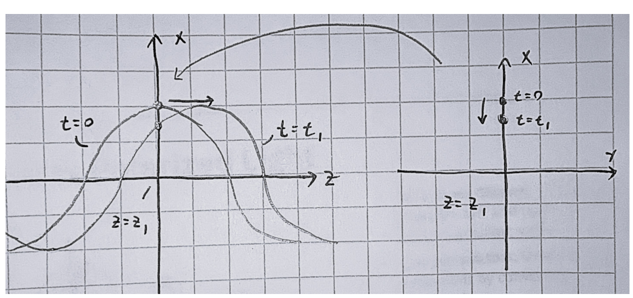
When we track a specific point on the wave, f must give the same value at different positions and times, meaning the phase must stay constant. This shows that as t increases, z must also increase to maintain the phase (the wave moves rightward!). Vice versa, as z increases, t must increase too to keep the phase constant (again, moving rightward!).

<u>Note:</u> This intuition applies to *any* point on the wave! Thus, by tracking all points, we can clearly see a propagating plane wave.

Local View

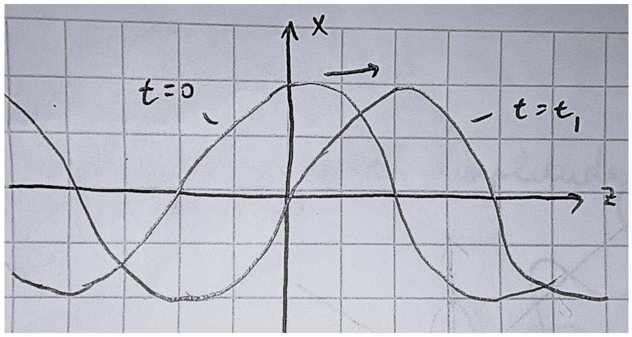
The second is a *local* view, looking at how the phase changes at a fixed position as time changes, or at a fixed time as position changes. For this, we will look at a sinusoidal wave, to better show oscillation: $\psi = cos(kz - \omega t)$

1. Fixing z, letting t vary:
   Imagine we are at a fixed position $z = z_1$. As time increases, we are effectively watching a cross-section of the wave at that point, seeing only a single point along the wave moving up and down like riding the "surface" of the wave.

2. Fixing t, letting z vary:
   Now imagine we are at a fixed time $t = t_1$. Moving along the z axis, and plotting the

wave's amplitude, we see the full shape of the wave at that instant (ωt_1 becomes a constant phase shift).



3. Letting z and t vary together:
   When we looked at z and t separately, we saw only a *part* of the wave's behavior. For a fixed z, we observed how a single point moves over time, capturing the wave's temporal variation. For a fixed t, we saw the spatial variation, revealing the full shape of the wave statically at that instant. When both z and t vary together, we can see the wave's complete dynamic behavior, watching how its shape propagates through space and time.

<u>Note:</u> This intuition will become very useful when we study electromagnetic polarization, where the plane wave is a 2D wave. In our cross-sectional view, the single point along the wave will move along the xy plane instead of just up and down.

Vector Calculus

- **Del Operator:**

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

- **Laplacian Operator:**

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \text{(in cartesian)}$$

- **Divergence:** Dot product of del operator with a vector field $\vec{A}$, giving a scalar field that represents how much $\vec{A}$ spreads (diverges) out from a point.

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

  <u>Note:</u> A positive divergence indicates a source, while a negative divergence indicates a sink.

- **Curl:** Cross product of del operator with a vector field $\vec{A}$, giving a vector field that represents how much $\vec{A}$ curls around a point.

$$\begin{aligned}
\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\
&= \underbrace{\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right)}_{\text{curl about } x\text{-axis}} \hat{x} + \underbrace{\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right)}_{\text{curl about } y\text{-axis}} \hat{y} + \underbrace{\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)}_{\text{curl about } z\text{-axis}} \hat{z}
\end{aligned}$$

<u>Note:</u> A positive curl indicates a counterclockwise rotation, while a negative curl indicates a clockwise rotation about the respective axis.

- **Vector Identities:**

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \quad \text{(Product Rule)}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad \text{(Curl of a curl rule)}$$

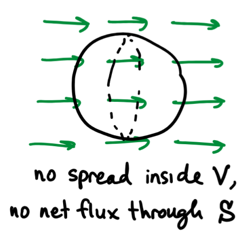
$$\vec{\nabla} \times (\vec{\nabla} v) = 0 \quad \text{(Curl of a gradient is always zero)}$$

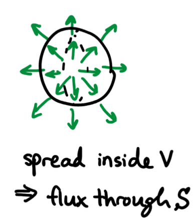
$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0 \quad \text{(Divergence of a curl is always zero)}$$

Divergence Theorem

Relates the flux of a vector field $\vec{F}$ through a closed surface S to the divergence of $\vec{F}$ within the volume V enclosed by S.

$$\underbrace{\int_V (\vec{\nabla} \cdot \vec{F})\, d\tau}_{\text{how field spreads inside the volume}} = \underbrace{\oint_S \vec{F} \cdot d\vec{a}}_{\text{flux of field through the surface}}$$

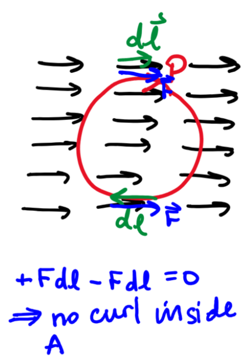


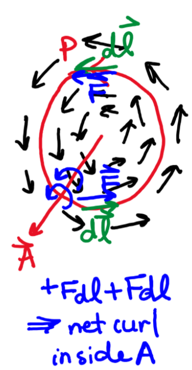


Stokes Theorem

States that inside a surface A bounded by a closed curve P, if a vector field $\vec{F}$ has a non-zero curl *about* $\vec{A}$ *(the normal to the surface)*, then there will be a non-zero circulation of $\vec{F}$ around the curve P.

$$\underbrace{\int_A (\vec{\nabla} \times \vec{F}) \cdot d\vec{a}}_{\text{flux of the curl of } \vec{F}} = \underbrace{\oint_P \vec{F} \cdot d\vec{l}}_{\text{circulation of } \vec{F}}$$





Key Electricity and Magnetism Laws

- **Coulomb's Law**

$$\vec{F}_e = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}$$

- **Electric Field of a Point Charge**
sets up a diverging field

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

- **Magnetic Force**

$$\vec{F}_b = q\vec{v} \times \vec{B}$$

- **Biot-Savart Law**
sets up a curling field

$$\vec{B} = \int \frac{\mu_0}{4\pi}\frac{i\, d\vec{s} \times \hat{r}}{r^2}$$

Unit 1: Electromagnetic Waves

Maxwell's Equations

1. **Gauss's Law for Electricity**
Relates the electric flux through a closed 3D Gaussian surface to the total charge enclosed within that surface.
$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\varepsilon_0}$$
There exists a diverging electric field if there exists a non-zero charge density at that point.
$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

2. **Gauss's Law for Magnetism**
Any 3D Gaussian surface will have zero net magnetic flux (no magnetic monopoles).
$$\oint_S \vec{B} \cdot d\vec{a} = 0$$
There exists no diverging magnetic field at any point in space because there are no magnetic monopoles.
$$\vec{\nabla} \cdot \vec{B} = 0$$

3. **Faraday's Law of Induction**
Relates the electric circulation around a closed Faradian loop to the changing magnetic flux through the surface bounded by the loop.
$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$
<u>Note:</u> For coils with N turns, multiply the flux by N.

A changing magnetic field induces a circulating (curling) electric field.
$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. **Ampere-Maxwell Law**
   Relates the magnetic circulation around a closed Amperian loop to the enclosed current and changing electric flux through the surface bounded by the loop.

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

   Both a changing electric field and a non-zero current density induce a circulating (curling) magnetic field.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}, \quad I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}$$

Deriving the Differential Forms of Maxwell's Equations

By applying the divergence and stokes theorems to the integral forms of Maxwell's equations, we can derive their corresponding differential forms. This process allows us to express the behavior of electric and magnetic fields locally at every point in space and time, rather than just over closed surfaces or loops.

1. **Gauss's Law for Electricity**

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

   Applying the divergence theorem to the left-hand side:

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{E})\, d\tau$$

   We can also write the enclosed charge as a volume integral of the charge density:

$$q_{\text{enc}} = \int_V \rho\, d\tau \quad \rho = (\text{vol charge density})$$

   Thus,

$$\implies \int_V (\vec{\nabla} \cdot \vec{E})\, d\tau = \frac{1}{\varepsilon_0} \int_V \rho\, d\tau$$

$$\implies \boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}}$$

2. **Gauss's Law for Magnetism**

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

   Applying the divergence theorem:

$$\oint_S \vec{B} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{B})\, d\tau = 0$$

   Thus,

$$\implies \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

3. **Faraday's Law of Induction**

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a}$$

We can also write the changing magnetic flux as:

$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left[\int_S \vec{B} \cdot d\vec{a}\right] = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

Note: Stokes theorem relates the circulation around a curve to the curl over a surface at a single instant in time. In other words, the surface S is *fixed* and does not change in time, so $d\vec{a}$ is *time independent* and the time derivative acts only on the integrand, allowing us to move the time derivative inside the integral.

Thus, equating the two sides gives:

$$\implies \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

$$\implies \boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

4. **Ampere-Maxwell Law**

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{B} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a}$$

We can also write the enclosed current as a surface integral of the current density:

$$I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}, \quad \vec{J} = \text{(area current density)}$$

And the changing electric flux as:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt}\left[\int_S \vec{E} \cdot d\vec{a}\right] = \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a}$$

Thus, equating the two sides gives:

$$\implies \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \int_S \left(\mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right) \cdot d\vec{a}$$

$$\implies \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

Applying Vector Identities
Additionally, we can apply the vector identities we have to the differential forms of Maxwell's equations to derive some useful results.

Applying divergence of a curl is zero to Faraday's law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) &= 0 \\ \vec{\nabla} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) &= 0 \\ -\frac{\partial}{\partial t} \underbrace{(\nabla \cdot \vec{B})}_{\text{Gauss's law for magnetism (=0)}} &= 0 \\ \implies 0 &= 0\end{aligned}$$

Applying divergence of a curl is zero to Ampere-Maxwell law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) &= 0 \\ \vec{\nabla} \cdot \left(\mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \underbrace{(\nabla \cdot \vec{E})}_{\text{Gauss's law for electricity}(=\frac{\rho}{\varepsilon_0})} &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\varepsilon_0} \right) &= 0\end{aligned}$$

$$\implies \boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} \quad \text{(Continuity Equation)}$$

We can rewrite the continuity equation into an integral form:

Integrating both sides over a volume V:

$$\int_V (\vec{\nabla} \cdot \vec{J}) \, d\tau = \int_V -\frac{\partial \rho}{\partial t} \, d\tau$$

Apply divergence theorem to the left-hand side:

$$\int_V (\vec{\nabla} \cdot \vec{J}) \, d\tau = \oint_S \vec{J} \cdot d\vec{a}$$

Since the volume V is fixed in space, we can pull the time derivative outside the integral. We also know that the total charge enclosed is $q = \int_V \rho \, d\tau$:

$$\int_V -\frac{\partial \rho}{\partial t} \, d\tau = -\frac{d}{dt} \underbrace{\int_V \rho \, d\tau}_{q_{\text{enclosed}}} = -\frac{dq}{dt}$$

Thus, equating the two sides gives:

$$\boxed{\oint_S \vec{J}\cdot d\vec{a} = -\frac{dq}{dt}}$$

Note: The continuity equation is also called a flux based conservation law. It represents a local **conservation of charge** and states that if the total charge enclosed in a volume is changing ($\frac{dq}{dt} \neq 0$), then there must be a net current flowing into or out of the volume ($\oint_S \vec{J}\cdot d\vec{a} \neq 0$) that accounts for that change.

If a balloon were filled with a certain amount of water and that quantity of water was decreasing...

It must mean that there is a flux of water passing through the surface, leaving the volume!

Showing that E and B Fields Exist as Waves in a Vacuum

In a vacuum, there are no charges or currents, so we can set $\rho = 0$ and $\vec{J} = \vec{0}$ in Maxwell's equations. This simplifies them to:

$$\vec{\nabla}\cdot\vec{E} = 0 \qquad \vec{\nabla}\cdot\vec{B} = 0 \qquad \text{no divergent fields (no source charge)}$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad \vec{\nabla}\times\vec{B} = \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} \qquad \text{only exist curling fields induced by one another!}$$

By applying the curl of a curl rule to Faraday's and Ampere-Maxwell laws, we can derive the wave equations for the electric and magnetic fields in a vacuum.

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}\underbrace{(\vec{\nabla}\cdot\vec{E})}_{\text{0 in a vacuum}} - \nabla^2\vec{E}$$

$$\vec{\nabla}\times\left(-\frac{\partial\vec{B}}{\partial t}\right) = -\nabla^2\vec{E}$$

$$-\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B}) = -\nabla^2\vec{E}$$

$$-\frac{\partial}{\partial t}\left(\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}\right) = -\nabla^2\vec{E}$$

$$\Longrightarrow \boxed{\nabla^2\vec{E} = \mu_0\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}}$$

$$\vec{\nabla}\times\underbrace{(\vec{\nabla}\times\vec{B})}_{\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}\text{ in a vacuum}} = \vec{\nabla}\underbrace{(\vec{\nabla}\cdot\vec{B})}_{0} - \nabla^2\vec{B}$$

$$\mu_0\varepsilon_0\frac{\partial}{\partial t}\underbrace{(\vec{\nabla}\times\vec{E})}_{-\frac{\partial\vec{B}}{\partial t}} = -\nabla^2\vec{B}$$

$$-\mu_0\varepsilon_0\frac{\partial^2\vec{B}}{\partial t^2} = -\nabla^2\vec{B}$$

$$\Longrightarrow \boxed{\nabla^2\vec{B} = \mu_0\varepsilon_0\frac{\partial^2\vec{B}}{\partial t^2}}$$

Recall: The classic wave equation is

$$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} \qquad \Longrightarrow \qquad v = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = c$$

Modeling EM Radiation (Light) in a Vacuum

Properties of light:

- $\vec{E} \times \vec{B}$ points in the direction of propagation.
- $\vec{E}$ and $\vec{B}$ are perpendicular to each other and to the direction of propagation (transverse wave).
- $\vec{E}$ and $\vec{B}$ oscillate in phase as to perpetually induce each other according to Faraday's and Ampere-Maxwell laws.
- The magnitudes of $\vec{E}$ and $\vec{B}$ are related by $E = cB$.
- Light does not require a medium to propagate.
- Light has speed
$$v = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Our goal is to find a solution for $\vec{E}$ and $\vec{B}$ that satisfies the 3D wave equations as well as Maxwell's equations in a vacuum.